

Customer_Churn_Retention_Analytics_Report

Project Overview:

AI-Powered Customer Churn & Retention Analytics is an end-to-end data science and analytics project designed to analyze customer churn in a telecommunications dataset. The project uses Python, SQL, Machine Learning, and Power BI to identify factors associated with customer churn, predict customer churn risk, estimate potential revenue at risk, and generate data-driven customer retention recommendations.

The project follows a complete analytics workflow, starting with data cleaning and exploratory analysis, followed by SQL-based business analysis, machine learning-based churn prediction, customer risk segmentation, and interactive Power BI visualization.

Business Problem:

Customer churn is a major business challenge for telecommunications companies because losing existing customers can reduce revenue and increase the cost of acquiring new customers.

The objective of this project is to analyze customer behavior and identify customers who are more likely to churn. By using historical customer data and machine learning, the project aims to estimate churn probability, segment customers according to risk, identify potential revenue at risk, and provide targeted retention recommendations.

The analysis can help businesses understand the factors associated with customer churn and support data-driven customer retention strategies.

Project Objective:

The main objectives of this project are:

To clean and preprocess the telecom customer dataset.

To perform exploratory data analysis and identify patterns related to customer churn.

To analyze customer churn using SQL and generate business insights.

To develop machine learning models for predicting customer churn.

To estimate the probability of churn for individual customers.

To segment customers into Low, Medium, and High Risk categories.

To estimate potential monthly revenue at risk.

To generate data-driven customer retention recommendations.

To develop an interactive Power BI dashboard for monitoring churn and customer risk.

To present the analysis in a form that can support business decision-making

Dataset Description:

The dataset used in this project is a telecommunications customer churn dataset containing customer demographic information, services, account details, billing information, and churn status.

The original dataset contains 7,043 customer records and 21 columns. During data cleaning, records with missing values in the TotalCharges field were removed, resulting in a cleaned dataset of 7,032 customer records.

The target variable is Churn, which indicates whether a customer left the company:

Yes — Customer churned

No — Customer did not churn

Important variables used in the analysis include:

customerID

gender

SeniorCitizen

Partner

Dependents

tenure

InternetService

OnlineSecurity

TechSupport

Contract

PaymentMethod

MonthlyCharges

TotalCharges

Churn

Data Cleaning & Processsing

The raw telecom customer dataset was cleaned and prepared before performing exploratory data analysis and machine learning.

The following preprocessing steps were performed:

Loaded the dataset using Pandas.

Checked the dataset structure, data types, and missing values.

Converted the TotalCharges column from text to numeric format.

Identified missing values in TotalCharges.

Removed records with missing TotalCharges, resulting in 7,032 usable customer records.

Converted the Churn target variable into numerical form, where Yes = 1 and No = 0 for machine learning.

Removed customerID from the machine learning features because it is an identifier and does not provide predictive information.

Converted categorical variables into numerical features using one-hot encoding.

Created additional features such as TenureGroup, CustomerValue, and MonthlyChargeGroup.

Prepared the processed data for exploratory analysis and machine learning.

Exploratory Data Analysis(EDA)

Exploratory Data Analysis was performed to understand customer characteristics and identify patterns associated with churn.

The analysis examined:

Overall customer churn distribution.

Churn rate by contract type.

Churn rate by tenure group.

Churn rate by internet service.

Churn rate by payment method.

Churn rate by online security.

Churn rate by technical support.

Relationship between monthly charges and churn.

Relationship between customer tenure and churn.

Customer value and its relationship with churn.

Correlations between important numerical variables.

The cleaned dataset contained 7,032 customers, with 1,869 churned customers and 5,163 customers who did not churn. The overall churn rate was 26.58%.

The analysis also showed that the average monthly charge was \$74.44 for churned customers compared with \$61.31 for customers who did not churn.

These findings were used to identify important customer characteristics and support the subsequent machine learning and retention analysis.

SQL Analysis

MySQL was used to perform structured analysis of the customer churn dataset and validate the findings obtained through Python.

The SQL analysis included:

Total number of customers.

Number of churned customers.

Overall churn rate.

Churn analysis by contract type.

Churn analysis by internet service.

Churn analysis by payment method.

Average monthly charges by churn status.

Churn rate by customer tenure.

Average total charges by churn status.

The SQL analysis confirmed that the cleaned dataset contains 7,032 customers and that the overall churn rate is 26.58%.

SQL results were also used to validate important findings from the Python analysis and provide business-oriented customer churn insights.

Machine Learning

Machine learning classification models were developed to predict whether a customer is likely to churn.

The Churn variable was used as the target variable. The dataset was divided into training and testing sets using an 80:20 split, with stratification to maintain the churn distribution.

The following machine learning models were trained and evaluated:

Logistic Regression

Decision Tree

Random Forest

XGBoost

The models were evaluated using:

Accuracy

Precision

Recall

F1-score

ROC-AUC

Confusion Matrix

The Random Forest model was subsequently used for customer churn probability and risk scoring in the project.

The trained Random Forest model was saved as:

churn_model.pkl

This model file is stored in the project's models folder and can be loaded again using Python without retraining the model.

Customer Risk Segmentation

After training the machine learning model, churn probabilities were generated for individual customers using the Random Forest model.

Customers were then divided into three risk categories using project-defined churn probability thresholds:

High Risk: Churn probability greater than 70%

Medium Risk: Churn probability from 40% to 70%

Low Risk: Churn probability below 40%

This segmentation helps prioritize customers for retention activities. High-risk customers can receive immediate attention, while medium-risk customers can be targeted with preventive retention campaigns.

The risk category was also used in the Power BI dashboard to analyze customer distribution, revenue at risk, and retention recommendations.

10.Revenue At Risk

Revenue at Risk was calculated to estimate the potential monthly revenue associated with customers who have a higher probability of churning.

The project uses the following formula:

Revenue at Risk = Monthly Charges × Churn Probability

This calculation combines the customer's monthly charges with the probability predicted by the machine learning model.

The resulting value was used to:

Identify customers with greater potential revenue exposure.

Prioritize high-risk customers for retention activities.

Analyze revenue exposure by customer risk category.

Create revenue-related KPIs and visualizations in Power BI.

Revenue at Risk is an analytical estimate based on predicted churn probability and should not be interpreted as guaranteed future revenue loss.

11.Retention Recommendation

Based on the predicted churn risk and selected customer characteristics, the project generates rule-based retention recommendations for at-risk customers.

Examples of recommendations include:

Offer long-term contract discounts to high-risk customers on month-to-month contracts.

Provide personalized pricing or discount offers to high-risk customers with higher monthly charges.

Promote technical support for high-risk customers without Tech Support.

Offer Online Security packages to high-risk customers without Online Security.

Use targeted retention promotions for medium-risk customers.

Continue regular engagement for low-risk customers.

These recommendations are generated using project-defined business rules and are intended as decision-support suggestions. They have not been experimentally validated as guaranteed interventions.

PowerBi Dashboard

An interactive Power BI dashboard was developed to present customer churn analysis, risk segmentation, revenue exposure, and retention recommendations in a business-friendly format.

The dashboard contains five main pages:

Executive Overview – Provides key KPIs and an overall view of customer churn.

Churn Analysis – Analyzes churn patterns across contract, tenure, internet service, payment method, online security, and technical support.

Customer Risk – Shows Low, Medium, and High Risk customer segments and revenue at risk.

High Risk Customers – Provides customer-level details for customers classified as High Risk.

Retention Recommendations – Displays recommended retention actions and their associated customer and revenue exposure.

The dashboard includes interactive filters, KPI cards, charts, tables, and navigation buttons to allow users to explore different aspects of customer churn.

Key dashboard metrics include:

Total Customers

Churned Customers

Churn Rate

Average Monthly Charges

Revenue at Risk

High Risk Customers

Medium Risk Customers

Low Risk Customers

The Power BI dashboard converts the analytical results into an interactive visual format that can support customer churn monitoring and retention analysis.

Key Business Insights

The analysis of the cleaned telecom customer dataset provided several important business insights:

The dataset contains 7,032 customers, of which 1,869 customers churned.

The overall customer churn rate was 26.58%.

Churned customers had higher average monthly charges (\$74.44) compared with customers who did not churn (\$61.31).

Customer churn was analyzed across different contract types, tenure groups, internet services, payment methods, online security, and technical support.

Machine learning was used to estimate the churn probability of individual customers.

Customers were segmented into Low, Medium, and High Risk categories to support prioritization.

Revenue at Risk was calculated to identify customers with greater potential monthly revenue exposure.

Rule-based retention recommendations were generated for at-risk customers.

The Power BI dashboard brings these findings together through interactive KPIs, visualizations, customer-level analysis, and retention recommendations.

These insights demonstrate how customer data, machine learning, and business intelligence tools can be combined to support data-driven churn and retention analysis.

Conclusion

The AI-Powered Customer Churn & Retention Analytics project demonstrates an end-to-end approach to analyzing customer churn using data science and business intelligence techniques.

The project combined Python, SQL, Machine Learning, and Power BI to clean and analyze customer data, identify churn-related patterns, predict customer churn probability, segment customers based on risk, estimate revenue at risk, and generate retention recommendations.

The analysis of 7,032 customers identified an overall churn rate of 26.58%. Machine learning models were developed to support customer risk prediction, while the Random Forest model was used to generate churn probabilities for individual customers.

The Power BI dashboard transformed these analytical results into interactive business insights, allowing users to explore churn patterns, customer risk, revenue exposure, and retention recommendations.

Overall, the project demonstrates how data analytics and machine learning can be applied to a real-world business problem and converted into actionable decision-support insights.

Future Improvement

The project can be further improved by adding the following features:

Streamlit Deployment: Deploy the churn prediction model as an interactive web application.

Hyperparameter Tuning: Optimize machine learning models to potentially improve predictive performance.

SHAP Explainability: Add detailed customer-level explanations showing why a customer is predicted to be at risk.

Automated Model Retraining: Implement a process for regularly updating the model with new customer data.

Database Integration: Connect Power BI directly to the MySQL database for easier data refresh.

Automated Retention Campaigns: Integrate risk predictions with customer communication systems.

Model Monitoring: Track model performance over time and detect changes in data patterns.

These improvements could make the project more suitable for a production-level customer churn and retention analytics system.

Project Structure & Technologies

The project is organized into separate folders for data, notebooks, SQL scripts, machine learning models, dashboards, and reports.

Project Structure:

```
Customer-Churn-Retention-Analytics/  
├──  
│   ├── data/  
│   │   ├── WA_Fn-UseC_-Telco-Customer-Churn.csv  
│   │   └── customer_churn_final.xlsx  
│   └──  
│       ├── notebooks/  
│       └── Customer_Churn_Analytics.ipynb
```

- - sql/
 - ■■■ database_creation.sql
 - ■■■ analysis_queries.sql
 - - dashboard/
 - ■■■ Customer_Churn_Retention_Analytics.pbix
 - - models/
 - ■■■ churn_model.pkl
 - - reports/
 - ■■■ Customer_Churn_Retention_Analytics_Report.pdf
 - - requirements.txt
 - README.md

Technologies Used:

Python

Pandas

NumPy

Matplotlib

Seaborn

Scikit-learn

XGBoost

SHAP

MySQL

Power BI

Jupyter Notebook

The combination of these technologies enabled the project to cover the complete workflow from raw customer data to machine learning predictions, business analysis, visualization, and retention decision support.